1. Let $\Omega \neq \emptyset$ and $\mathcal{P}(\Omega)$ is the collection of all subsets of Ω, i.e. the power set. Let $\mathcal{G}, \mathcal{C}, \mathcal{F} \subset \mathcal{P}(\Omega)$ be a classes of subsets of Ω, and $\lambda,\ \mu$ be set functions.

   (a) Define when we say $\mathcal{C}$ is a (i) semi-algebra, (ii) algebra (iii) σ-algebra(15 pt, 3×5)

(b) Suppose $\mathcal{G} \neq \emptyset$ is a semi-algebra, and also closed under finite *disjoint* unions, i.e. if $A \cap B = \emptyset$ and $A, B \in \mathcal{G}$, then $A \cup B \in \mathcal{G}$. Prove $\mathcal{G}$ is an algebra. (15 pts)

(c) Suppose $\mathcal{F} \neq \emptyset$ is a semi-algebra and is closed under finite disjoint unions. In addition, it is also a *monotone* class: i.e.
if $\{A_i\} \in \mathcal{F},\ A_i \uparrow$, then $\cup A_i \in \mathcal{F}$ and if $\{B_i\} \in \mathcal{F},\ B_i \downarrow$, then $\cap B_i \in \mathcal{F}$. Prove $\mathcal{F}$ is a σ-algebra. (10 pts)

(d) Suppose λ be a set function on the measurable space $(\Omega,\ \mathcal{F})$, where $\mathcal{F}$ is a σ-algebra. Let λ satisfy $\lambda : \mathcal{F} \to [0, \infty]$; $\lambda(\emptyset) = 0$, and it is *finitely additive* and *countably sub-additive:* i.e. if $A \cap B\ \emptyset$ and $A, B \in \mathcal{F}$, then $\lambda(A \cup B) = \lambda(A) + \lambda(B)$; and if $\{A_i\} \subset \mathcal{F}$ then $\lambda(\cup A_i) \leq \sum_i \lambda(A_i)$. Prove that λ is a measure on $(\Omega, \mathcal{F})$ (15 pts)

2. Let $F(x)$ be the cdf of standard normal (Gaussian) distribution, i.e.

$$F(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du, \ x \in \mathbb{R}$$

We know that the integrand is symmetric around 0 and the integral does **not** allow a closed-form. Instead, the cdf is numerically computed for intervals and presented in the standard normal table (the so called z-table) part of which is given below. Figure 1: Part of Z-table. This shows, for example, F (1.07) = 0.8577.

Standard Normal Probabilities

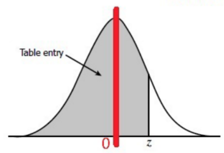

Table entry for z is the area under the standard normal curve to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	[illegible]	[illegible]	.8686	.8708	.8729	.8749	.8770	[illegible]	[illegible]	.8830
1.2	[illegible]	[illegible]	.8888	.8907	[illegible]	[illegible]	8962	[illegible]	[illegible]	[illegible]

(a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. [*Write the steps of the theorems carefully, stating the results used in each step*] (15 pt)

(b) Why is such a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ unique? (10 pts)

(c) For this extended measure μ_F^* (or μ_F), find $\mu_F^*(\{x \in \mathbb{R} : |x| \geq 0.5\})$ (10 pts)

(d) Let m be the Lebesgue measure (i.e. Lebesgue Steiljes measure with $F(x) = x$. We know already that m is translation and reflection invariant. In addition, it satisfies the scaling property:

$$m(cA) = |c|m(A),\ c \in \mathbb{R}, A \in \mathcal{B}(\mathbb{R}).$$

Prove this scaling property.